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In []:

Write a Python program using NumPy to analyze your daily step count data collected over the past month. The program should:

- Compute key statistics such as total steps, average steps per day, and maximum/minimum step counts.
- Identify days with high activity (e.g., days with steps above a certain threshold).
- Analyze trends over time, such as increases or decreases in activity, and optionally visualize them.

In [1]:

```
import numpy as np

np.random.seed(42)
grades = np.random.randint(50, 101, (30, 3))

math_avg, sci_avg, eng_avg = np.mean(grades, axis=0)
math_max, sci_max, eng_max = np.max(grades, axis=0)

print(f"Math: Avg {math_avg:.1f}, High {math_max}")
print(f"Science: Avg {sci_avg:.1f}, High {sci_max}")
print(f"English: Avg {eng_avg:.1f}, High {eng_max}")

total_avg = np.mean(grades)
print(f"\nOverall Class Average: {total_avg:.1f}")

total_scores = np.sum(grades, axis=1)

best_score = np.max(total_scores)
best_student_id = np.argmax(total_scores)

print(f"Top Student is ID #{best_student_id} with {best_score} total points.")
```

Math: Avg 76.6, High 100

Science: Avg 72.8, High 100

English: Avg 73.5, High 98

Overall Class Average: 74.3

Top Student is ID #21 with 273 total points.

